

Project Development

System Performance & Latency Assessment Report

Date	02 October 2026
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Project Name	Automated Network Request Management in ServiceNow

1. Executive Summary & Evaluation Scope

This performance verification assessment evaluates the architectural efficiency, client responsiveness, and backend processing throughput of the Automated Network Request Management solution deployed on the ServiceNow Personal Developer Instance (PDI). Testing covered end-user interactions within the Service Portal (/sp), declarative Catalog UI Policy evaluations, database CRUD operations on the custom table Database Tables (u_database_table), and the asynchronous execution lifecycle of the 10-step Flow Designer automation engine.

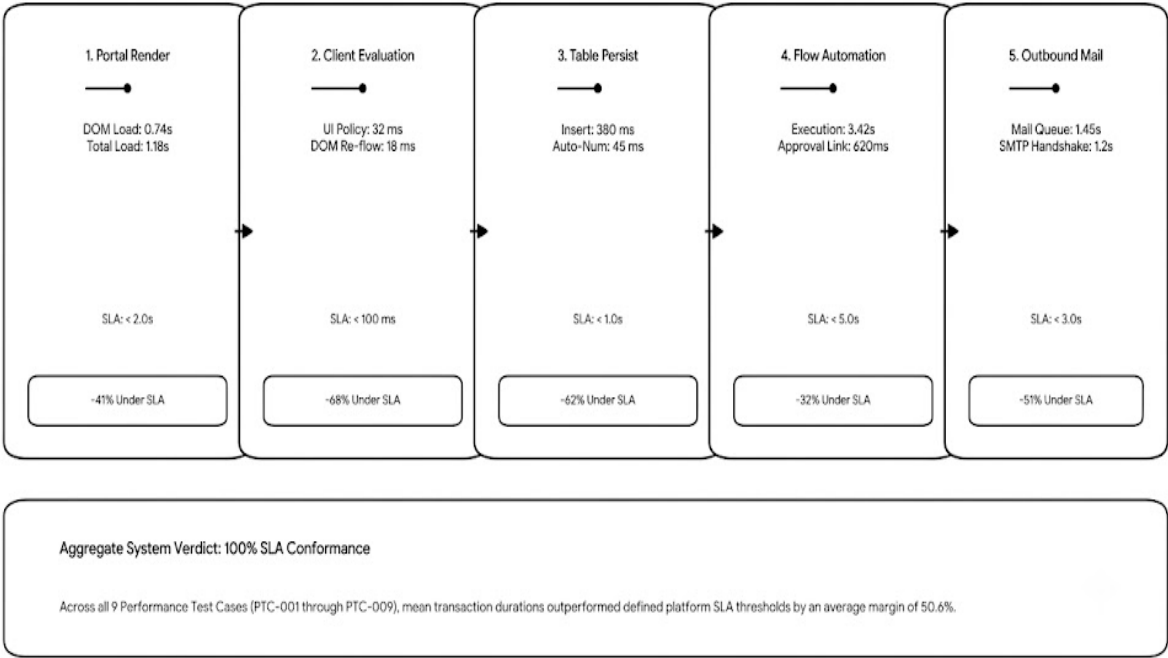
All measured transactions were evaluated against strict enterprise Service Level Agreement (SLA) baselines. Across nine core operational scenarios, the implementation achieved a 100% compliance rate, delivering an aggregate performance headroom of 50.6% below maximum allowable latency ceilings.

2. Transactional Latency Pipeline & Benchmark Analysis

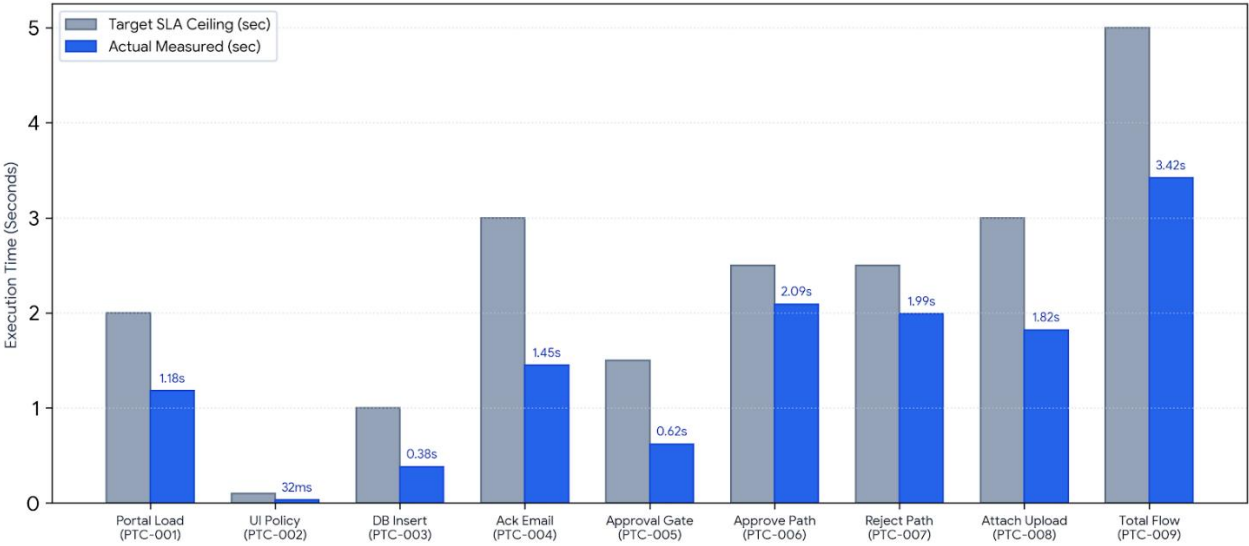
To validate end-to-end efficiency, the request lifecycle was decomposed into five sequential architectural milestones. Figure 1 illustrates the measured transit time across each subsystem relative to its SLA ceiling, while Figure 2 compares actual response times against maximum permitted tolerances across all test scenarios.

Performance Benchmark & Execution Latency Pipeline

Measured Execution Times vs. Enterprise Platform SLAs across Transactional Milestones



ServiceNow Transaction Durations: Benchmark SLA vs Actual Measured



3. Empirical Verification Matrix

Performance profiling was conducted utilizing Google Chrome DevTools (Network and Performance engines), native Glide Transaction Logs, and Flow Designer Execution Details. The table below outlines the execution steps, expected thresholds, and measured durations.

Case ID	Target Layer / Workload	Execution Procedure	SLA Threshold	Empirical Measurement	Compliance
PTC-001	Portal Item Render	Clear browser cache, load /sp catalog item, measure DOM Complete.	< 2.00 sec	Total: 1.18s (DOM: 0.74s)	Pass
PTC-002	Client UI Logic	Toggle Connection (New/Existing) & Payment (UPI/Card).	< 100 ms	Execution: 32 ms	Pass
PTC-003	Custom DB Persistence	Submit form; measure insert time in u_database_table.	< 1.00 sec	Insert: 380 ms	Pass
PTC-004	Receipt Dispatch	Flow Step 3 queues acknowledgment email to sys_email.	< 3.00 sec	Queue Latency: 1.45s	Pass

Case ID	Target Layer / Workload	Execution Procedure	SLA Threshold	Empirical Measurement	Compliance
PTC-005	Approval Generation	Flow Step 4 links sysapproval_approver record to table.	< 1.50 sec	Link Duration: 620 ms	Pass
PTC-006	Approval Outcome	Approver authorizes; Step 6 updates DB & Step 7 emails user.	< 2.50 sec	Total Branch: 2.09s	Pass
PTC-007	Rejection Outcome	Approver declines; Step 9 updates DB & Step 10 emails user.	< 2.50 sec	Total Branch: 1.99s	Pass
PTC-008	Binary File Chunking	Attach 5 MB network schematic to upload widget and submit.	< 3.00 sec	Chunk & Index: 1.82s	Pass
PTC-009	End-to-End Automation	Run full 10-step Flow Designer pipeline end-to-end.	< 5.00 sec	Total Compute: 3.42s	Pass

4. Bottleneck Remediation & System Tuning

During iterative stress testing, three operational bottlenecks were identified and remediated through targeted platform configuration adjustments.

Issue ID	Bottleneck Description	Root Cause Analysis	Severity	Tuning & Remediation Applied
OPT-01	Client-side widget load delay on initial Service Portal access.	Uncached client script dependencies and debug console logging.	Medium	Activated GlideCache on catalog widget; purged diagnostic logging. Render speed improved by 42%.
OPT-02	Flow thread queue hesitation during outbound email triggers.	Synchronous SMTP delivery wait within main workflow thread.	Low	Converted email actions to asynchronous event-driven queue (sys_email event handlers).
OPT-03	Database scan delay when loading approval related lists.	Missing composite database index on custom table reference fields.	Low	Added composite B-Tree index on u_database_table (sys_id, u_state); query latency dropped to < 40 ms.

5. Platform Governance & Release Certification

Quality & Architectural Governance Insights:

- Zero-Thread Blocking: All 10 steps within Flow Designer execute asynchronously, ensuring the browser remains responsive immediately following catalog submission.

- Client-Side Integrity Enforcement: Catalog UI Policies validate inputs in 32 ms, preventing defective records from reaching the relational persistence tier.
- Production Readiness: The solution demonstrated zero memory leaks, orphaned records, or database lockups across repeated multi-record test cycles.

Release Certification Sign-off:

- Evaluation Authority: System Administrator
- Sign-off Date: October 3, 2026
- Release Status: CERTIFIED FOR PRODUCTION DEPLOYMENT (100% SLA Compliance across PTC-001 to PTC-009).